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import numpy as np
import matplotlib.pyplot as plt

# --- 1. Problem Definition ---
# Calculate the entropy change for 1 mole of an ideal gas
↳ expanding
# from an initial state (T1, V1) to a final state (T2, V2).
#
# Assumptions:
# - Ideal Gas behavior (PV = nRT)
# - Constant Heat Capacity (Cv)

# Constants
R = 8.314 # Universal Gas Constant (J/mol*K)

# User-Defined Parameters
n = 1.0 # Moles of gas
Cv = 1.5 * R # Heat capacity at constant volume (monatomic
↳ ideal gas)
T1 = 300.0 # Initial Temperature (K)
V1 = 0.01 # Initial Volume (m^3)
T2 = 300.0 # Final Temperature (K) (Isothermal expansion for
↳ this example)
V2 = 0.02 # Final Volume (m^3)

# --- 2. Calculation ---
# The formula for entropy change of an ideal gas (molar basis)
↳ is:
# Delta_S = Cv * ln(T2/T1) + R * ln(V2/V1)

delta_s_molar = Cv * np.log(T2 / T1) + R * np.log(V2 / V1)
delta_s_total = n * delta_s_molar

# --- 3. Output Results ---
print(f"--- Process Conditions ---")
print(f"Initial State: T1 = {T1} K, V1 = {V1} m^3")
print(f"Final State: T2 = {T2} K, V2 = {V2} m^3")
print(f"Gas Properties: n = {n} mol, Cv = {Cv:.2f} J/mol*K")
print(f"\n--- Entropy Calculation ---")
print(f"Change in Entropy (Delta S): {delta_s_total:.4f} J/K")

# --- 4. Visualization (P-V Diagram) ---
# Create an array of volumes from V1 to V2 for plotting
v_vals = np.linspace(V1, V2, 100)

# Calculate Pressure for an Isothermal process (P = nRT/V)
# Note: If T changed, we would need to define the path (e.g.,
↳ adiabatic).

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# Here we assume T is constant for the plot path just for
↪ visualization.
p_vals = (n * R * T1) / v_vals

plt.figure(figsize=(8, 6))
plt.plot(v_vals, p_vals, 'b-', linewidth=2, label='Expansion
↪ Path (Isothermal)')
plt.plot(V1, (n*R*T1)/V1, 'ro', label='State 1 (Initial)')
plt.plot(V2, (n*R*T2)/V2, 'ko', label='State 2 (Final)')
plt.fill_between(v_vals, p_vals, color='skyblue', alpha=0.3,
↪ label='Work Done')

plt.xlabel('Volume ($m^3$)')
plt.ylabel('Pressure (Pa)')
plt.title(f'P-V Diagram for Gas Expansion\n$\Delta S =
↪ \{delta\_s\_total:.2f\}$ J/K')
plt.legend()
plt.grid(True)
plt.show()

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--- Process Conditions ---

Initial State: $T_1 = 300.0 \text{ K}$, $V_1 = 0.01 \text{ m}^3$

Final State: $T_2 = 300.0 \text{ K}$, $V_2 = 0.02 \text{ m}^3$

Gas Properties: $n = 1.0 \text{ mol}$, $C_v = 12.47 \text{ J/mol}\cdot\text{K}$

--- Entropy Calculation ---

Change in Entropy (ΔS): 5.7628 J/K

